

Diagnosis of Immediate Hypersensitivity

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Make.
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Learning Objectives

By the end of this lecture, the student will be able to

1. Describe the difference between different types of hypersensitivity reactions
2. Describe some of the physical exam findings in immediate hypersensitivity reactions
3. Describe some of the physical exam findings in atopy
4. Select appropriate laboratory diagnostic tests for evaluation of patients with hypersensitivity reactions
5. Select or identify appropriate treatments for immediate hypersensitivity reactions

Case Study: Jack

A 4-year-old boy named Jack has been brought into the ED with swelling of the face, generalized itchiness, and drowsiness. The symptoms developed a few minutes after eating a chocolate bar. His mother explains that he has had minor reactions to pistachios in the past, including lip swelling. The boy also has eczema, and his mother thinks that in the past, this has flared up when he has eaten hazelnuts. She has concluded that he might be allergic to tree nuts and has tried to eliminate them from the house. The patient has never had any symptoms from milk. Jack asks if you're going to take away the itch, and if he can have more chocolate when he gets home.

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Immunology for Medical Students, 3rd Ed.



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What is Hypersensitivity?

- An exaggerated or inappropriate immune response to benign antigens.
- The **immune response**, **NOT THE ANTIGEN**, is harmful to the body.
- First contact with the antigen sensitizes/primes the adaptive immune system.
- Subsequent contact(s) elicit hypersensitive response.

ABCD

Type I: Anaphylaxis/Atopic

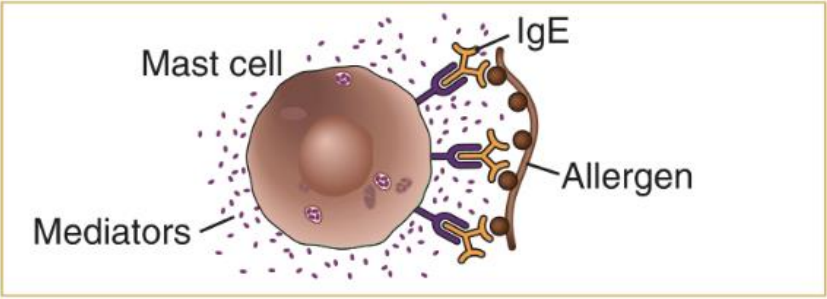
Type II: Antibody-mediated

Type III: immune Complex

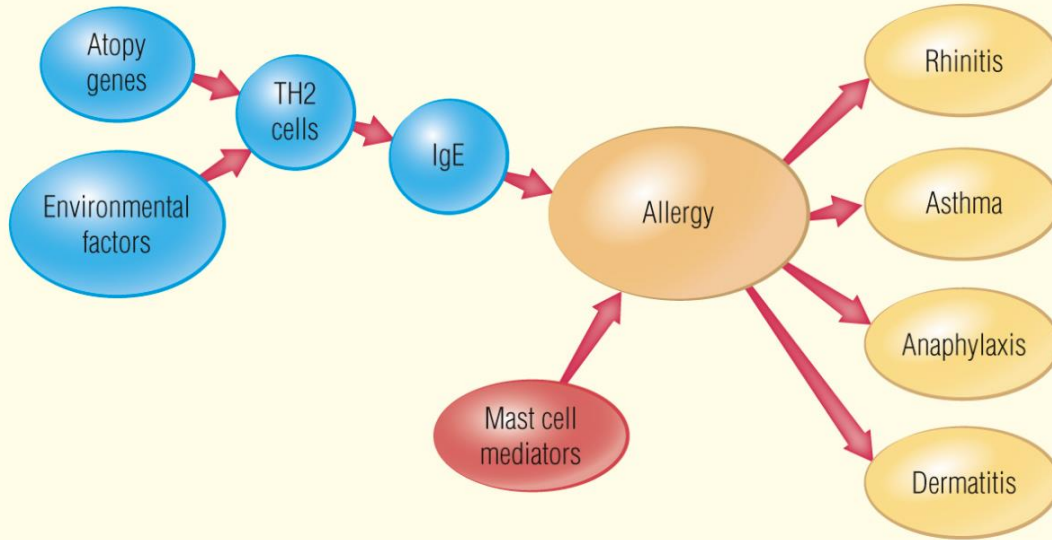
Type IV: Delayed

Type I Hypersensitivity

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Type of hypersensitivity	Pathologic immune mechanisms	Mechanisms of tissue injury and disease
Immediate hypersensitivity (Type I)	Th2 cells, IgE antibody, mast cells, eosinophils 	Mast cell-derived mediators (vasoactive amines, lipid mediators, cytokines) Cytokine-mediated inflammation (eosinophils, neutrophils)

Type I Hypersensitivity



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Biphasic Reaction

- Acute phase: minutes after antigen exposure
- Late phase: 2-24 hours after antigen exposure

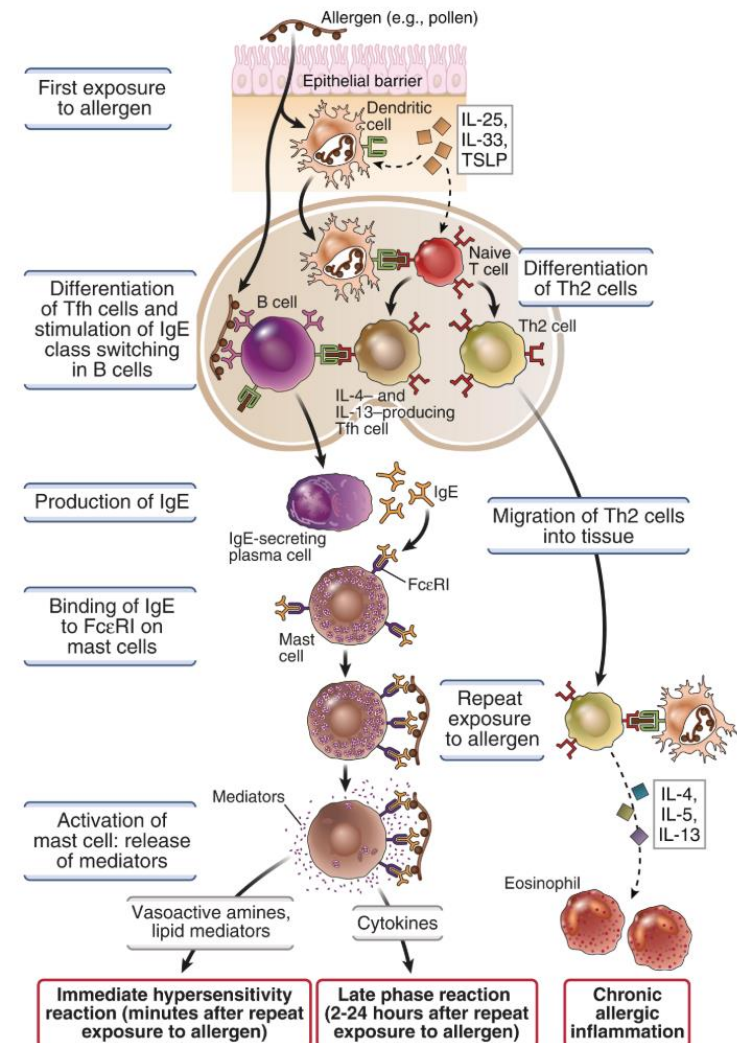


FIG. 11.2

Basic Immunology: Functions and Disorders of the Immune System, 11th Ed.

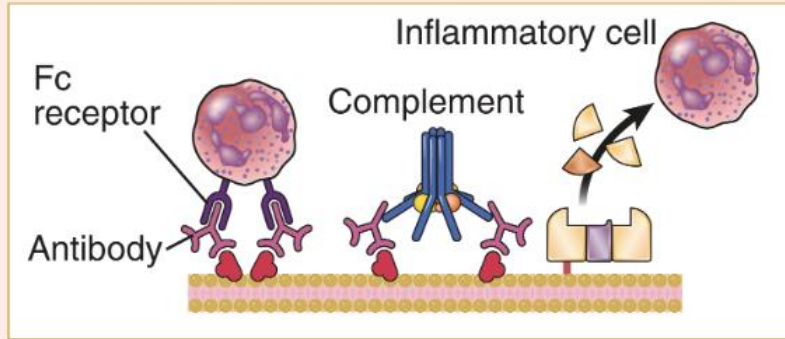
Type II Hypersensitivity

Type of hypersensitivity

Antibody-mediated diseases (Type II)

Pathologic immune mechanisms

IgM, IgG antibodies against cell surface or extracellular matrix antigens



Mechanisms of tissue injury and disease

Complement- and Fc receptor-mediated recruitment and activation of leukocytes (neutrophils, macrophages)

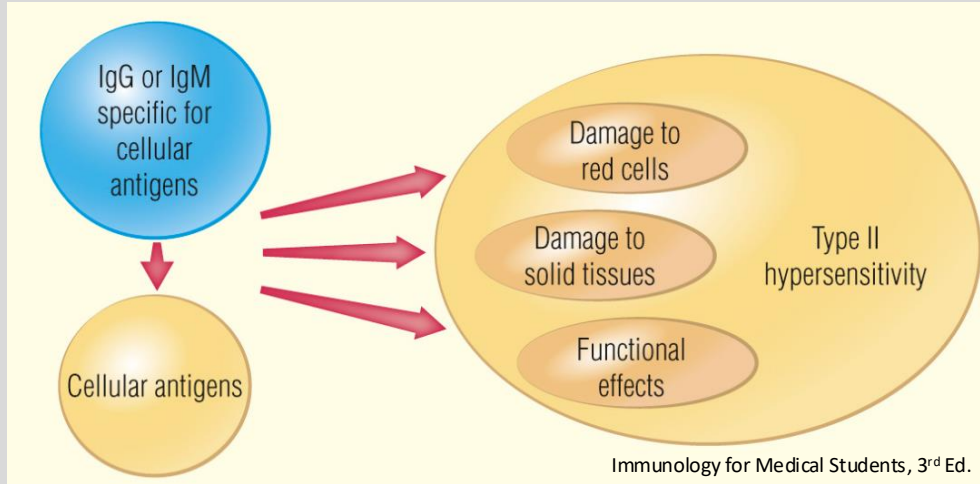
Opsonization and phagocytosis of cells

Abnormalities in cellular function, e.g. hormone receptor signaling

Basic Immunology: Functions and Disorders of the Immune System, 11th Ed.

- **Cellular destruction:** opsonization leading to phagocytosis or natural killer cell activation
- **Inflammation:** activation of complement system and Fc receptor-mediated inflammation
- **Cellular dysfunction:** abnormal blockade or activation of downstream process

Type II Hypersensitivity

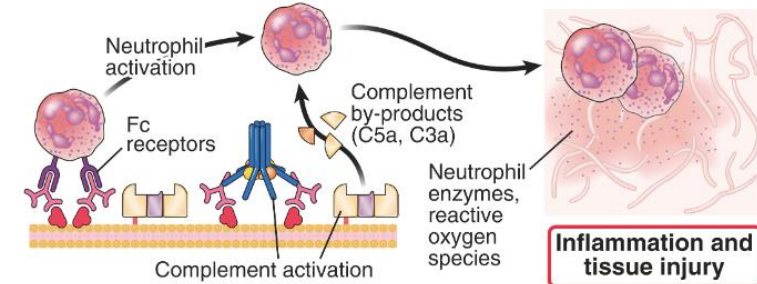


Cellular destruction: autoimmune hemolytic anemia, transfusion reaction, hemolytic disease of newborn

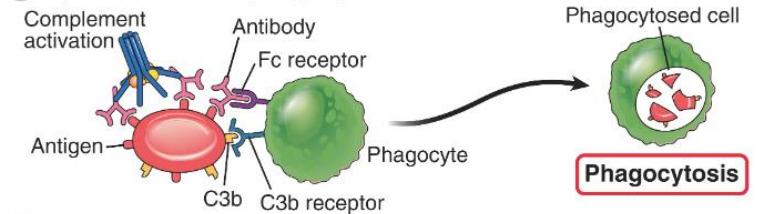
Inflammation: rheumatic fever, hyperacute transplant rejection, Goodpasture syndrome

Cellular dysfunction: Myasthenia gravis, Graves disease

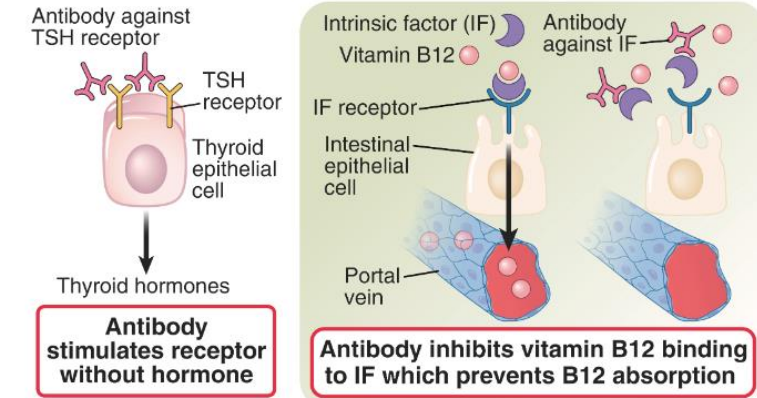
A Complement- and Fc receptor-mediated inflammation



B Opsonization and phagocytosis



C Abnormal physiologic responses without cell/tissue injury



Type III Hypersensitivity

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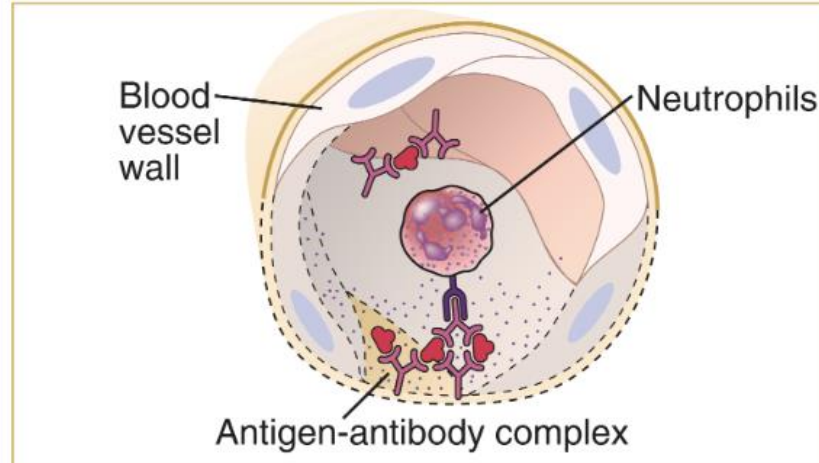
Type of
hypersensitivity

Pathologic immune mechanisms

Mechanisms of tissue
injury and disease

Immune
complex-
mediated
diseases
(Type III)

Immune complexes of circulating antigens
and IgM or IgG antibodies deposited in
vascular basement membrane



Complement- and
Fc receptor-mediated
recruitment and
activation of leukocytes

Type III Hypersensitivity

Serum sickness: **systemic** inflammatory disease with fever, rash, arthritis

- Production of antibodies against foreign antigen → 1-2 weeks later immune complexes form and deposit in tissues

Arthus reaction: **local** vasculitis

- Intradermal injection of antigen into pre-sensitized individual leads to immune complex formation in skin

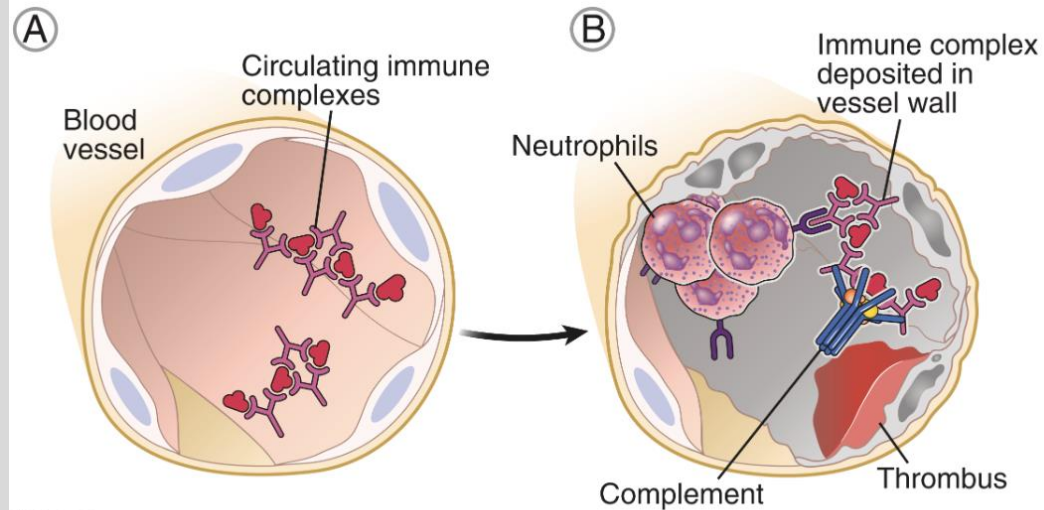
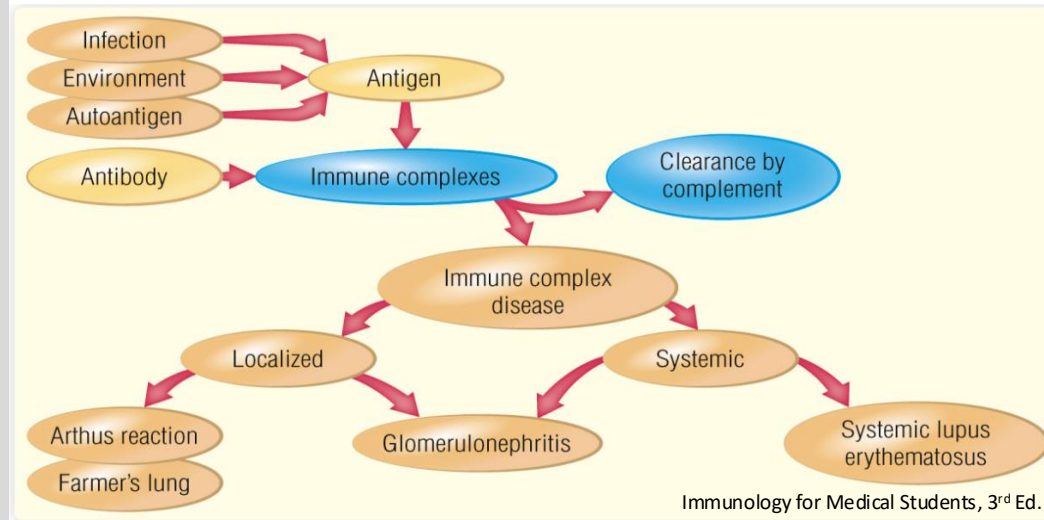
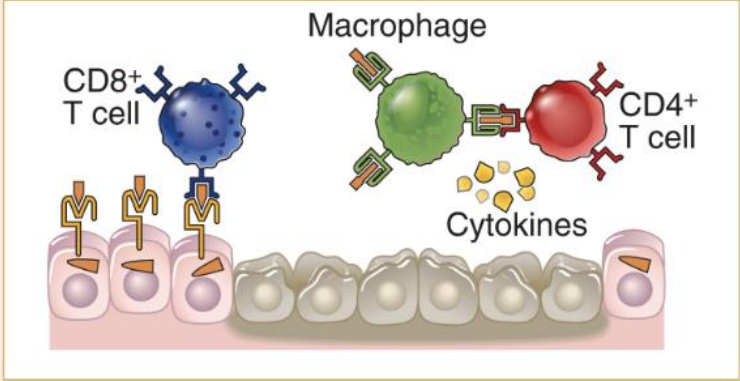


FIG. 11.9

Basic Immunology: Functions and Disorders of the Immune System, 11th Ed



Type IV Hypersensitivity

Type of hypersensitivity	Pathologic immune mechanisms	Mechanisms of tissue injury and disease
T cell-mediated diseases (Type IV)	1. Cytokine-mediated inflammation (CD4⁺ T cells) 2. T cell-mediated killing (CD8⁺ CTLs)  The diagram illustrates the cellular interactions in Type IV hypersensitivity. At the bottom, a layer of epithelial cells is shown. Above them, a CD8⁺ T cell (blue) and a CD4⁺ T cell (red) are interacting with a Macrophage (green). The CD8⁺ T cell is releasing cytotoxic granules (orange) towards the epithelial cells. The CD4⁺ T cell is releasing cytokines (yellow dots) which are acting on the Macrophage. The Macrophage is also releasing cytokines (yellow dots) which are acting on the CD4⁺ T cell. The CD4⁺ T cell is also releasing cytokines (yellow dots) which are acting on the epithelial cells.	1. Macrophage activation, cytokine-mediated inflammation 2. Direct target cell lysis, cytokine-mediated inflammation

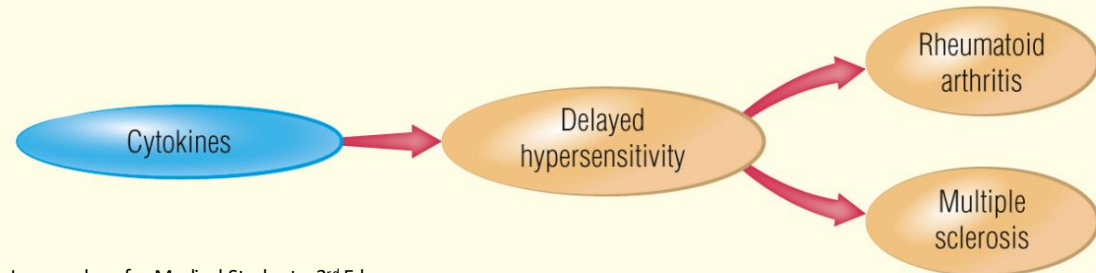
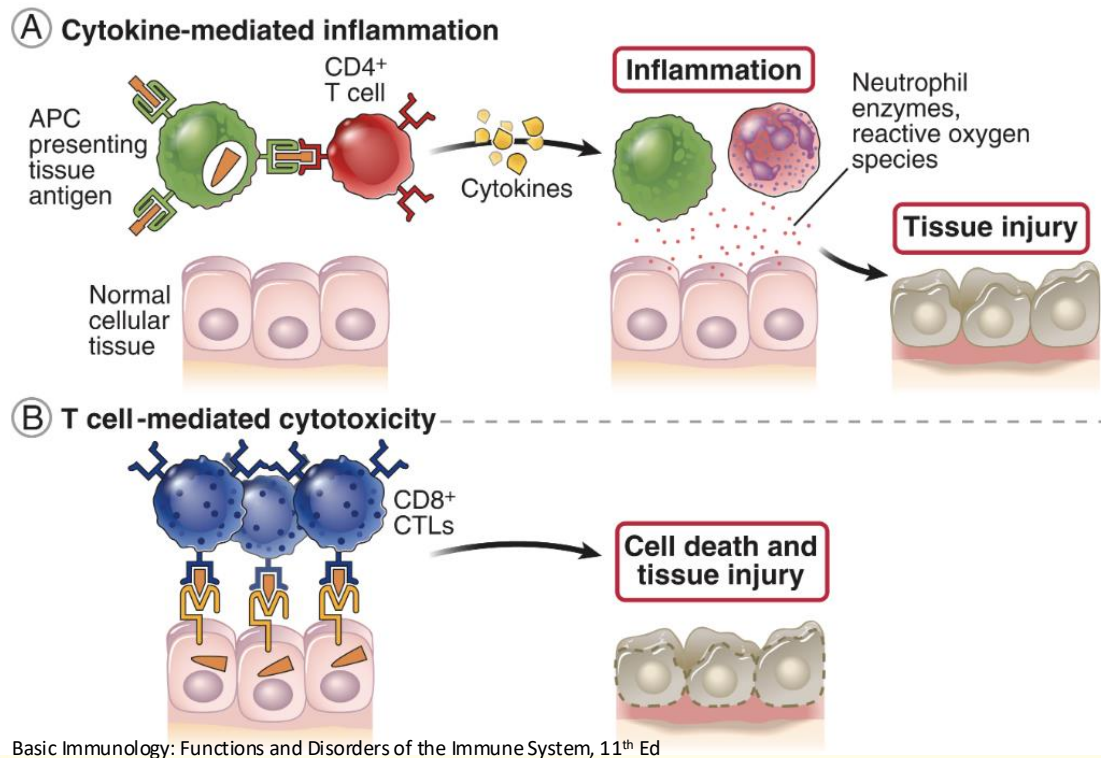
Basic Immunology: Functions and Disorders of the Immune System, 11th Ed

- **Direct cell cytotoxicity:** CD8⁺ cytotoxic T cells kill targeted cells
- **Inflammatory reaction:** effector CD4⁺ T cells recognize antigen and release inflammation-inducing cytokines

Type IV Hypersensitivity

Occurs **24 to 48 hours** after antigen exposure/challenge

4T's: T cells, Transplant rejections, Tuberculosis skin test, Touching (contact dermatitis)



ABCD

Type I: Anaphylaxis/Atopic

Type II: Antibody-mediated

Type III: immune Complex

Type IV: Delayed

Immediate Hypersensitivity: Anaphylaxis

1. NO known allergen exposure + sudden onset illness with skin/mucosal involvement + either respiratory involvement or cardiovascular involvement
2. Likely or known allergen exposure + sudden onset of TWO or more of the following: skin/mucosal involvement, respiratory involvement, cardiovascular involvement, severe GI involvement
3. Known allergen exposure + sudden onset of EITHER respiratory involvement after exposure to a non-inhaled allergen OR cardiovascular involvement

Immediate Hypersensitivity: Anaphylaxis

Skin/Mucosal: urticaria, flushing, erythema, facial/lip/tongue/oropharyngeal swelling or itching, difficulty swallowing, mottling (infants), repetitive lip licking (infants)

Respiratory: wheezing, increased work of breathing, hypoxemia, cough, dyspnea, stridor, voice change, hoarse cry (infants)

Cardiovascular : hypotension, syncope, dizziness, altered mental status, persistent unexplained tachycardia (infants)

Severe GI: crampy abdominal pain, repetitive vomiting, diarrhea

Immediate Hypersensitivity: Anaphylaxis

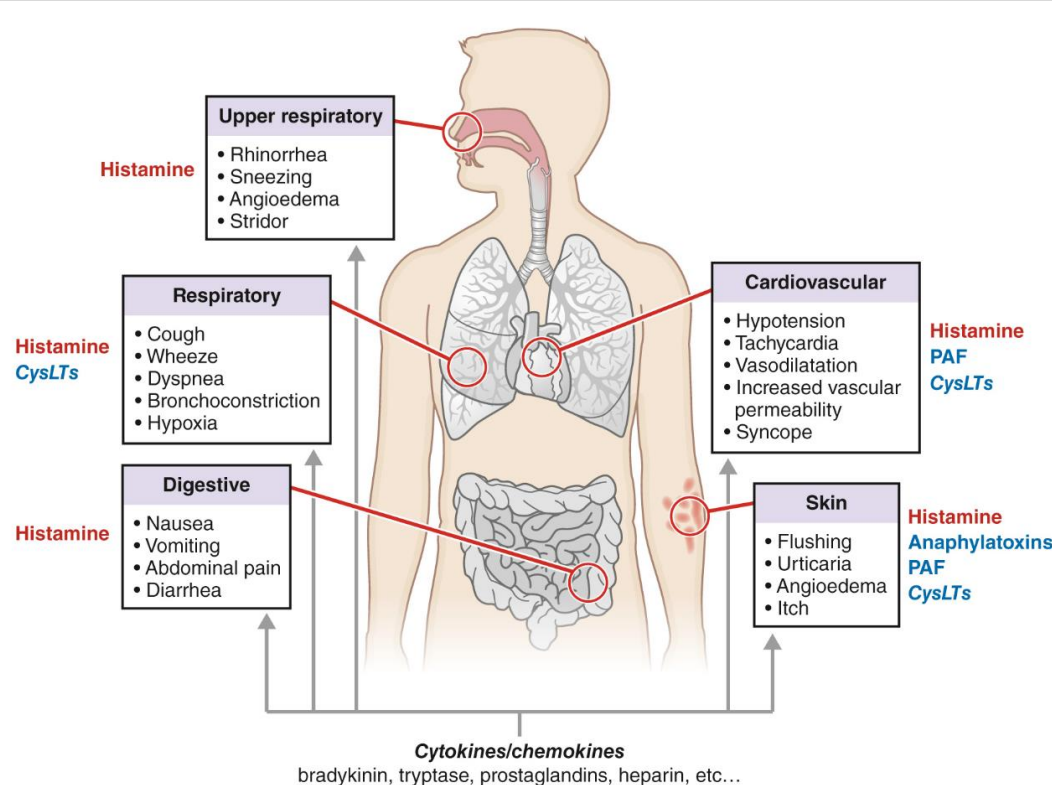


FIG. 75.3

Etiologies:

- Medications
- Foods
- Animal/Human Proteins
- Insect Venoms
- Vaccines
- Latex Products
- IV Contrast Media/Dyes/Colorants
- Enzymes
- Hormones
- Allergy Testing
- Allergen-Specific Immunotherapy
- Idiopathic

Immediate Hypersensitivity: Anaphylaxis

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Angioedema



Urticaria



Anaphylaxis Management

1. Remove exposure to the allergen.
2. Assess ABCs (airway, breathing, circulation) + mentation AND
2. Administer epinephrine intramuscularly to anterolateral thigh (vastus lateralis m.) or intranasally.
3. CALL 9-1-1. **ALL PERSONS WHO HAVE BEEN ADMINISTERED EPINEPHRINE MUST GO TO THE ED.**
4. If ABCs are acceptable, place the patient in the supine position and elevate lower extremities.
5. CONTINUOUSLY EVALUATE ABCs.



Anaphylaxis Management

- Epinephrine: 0.01 mg/kg/dose of the 1 mg/mL concentration with maximum dose of 0.3 mg in the prepubertal child and 0.5 mg in the adolescent.
- Epinephrine Auto-Injector (EAI):
 - 7.5 to less than 15 kg = 0.1 mg.
 - 15 to less than 30 kg = 0.15 mg.
 - 30 kg or more = 0.3 mg.
- Epinephrine Intranasal (NEFFY):
 - 15kg to less than 30kg = 1 spray of 1mg.
 - 30kg or more = 1 spray of 2mg.



Anaphylaxis Management

Second-Line Pharmacotherapy

- Second-generation H1 anti-histamine
 - Cetirizine (preferred)
 - Diphenhydramine
- H2 anti-histamine (relief from hives?)
 - Famotidine, ranitidine
- Corticosteroids: DO NOT PREVENT BIPHASIC REACTIONS
 - Methylprednisolone, prednisolone/prednisone



Anaphylaxis Management

Adjuvant Measures/Pharmacotherapy

- Nebulized albuterol for refractory bronchospasm
- Isotonic fluid bolus 20 mg/kg + Epinephrine infusion (0.01 mcg/kg/min) + Trendelenberg positioning for hypotension
- Vasopressors for refractory hypotension
- Glucagon for refractory hypotension
- Supplemental oxygen for hypoxia



Anaphylaxis Management

When to Observe for Prolonged Period (8-24h)/Admit:

1. Moderate to severe reaction (hypoxemia, airway edema, shock)
2. Required more than 1 dose of epinephrine
3. Prolonged duration of symptoms (>4 hours)
4. History of severe asthma or wheezing
5. History of severe or biphasic reactions
6. Ingested antigen with possibility of continued absorption

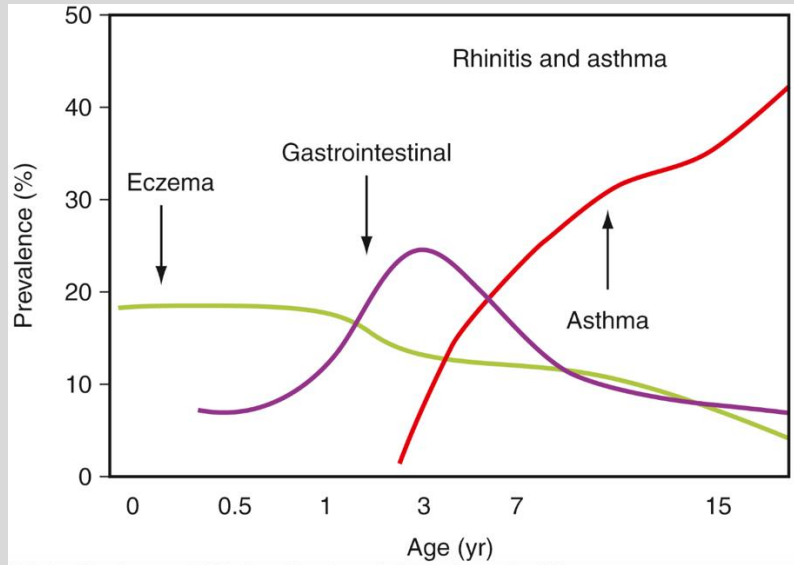
DISCHARGE PATIENT WITH EPINEPRHINE FOR SELF-ADMINISTRATION

- Provide the patient with an anaphylaxis action plan
- Provide education on allergen avoidance, hidden allergens, and cross-reactivity
- Instruct the patient to wear or carry identification of condition
- Refer to a board-certified allergist for identification of etiology and further education

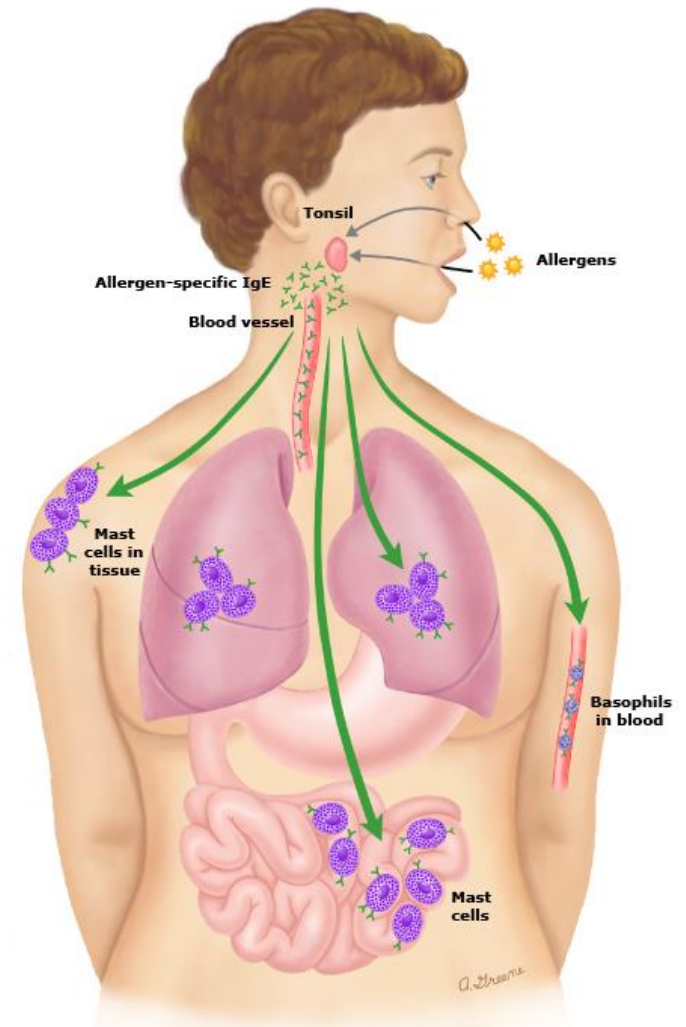
What is Atopy?

“Allergic triad”/“Allergic march”/“Atopic march”

Atopic dermatitis in infancy **PROGRESSES** TO childhood/young adult asthma + allergic rhinitis



Henry's Clinical Diagnosis and Management by Laboratory Methods, 24th Ed.



The relationship between IgE and allergic disease, UpToDate

Atopic Dermatitis

Chronic, **pruritic**, inflammatory skin disease

Acute: erythematous papules and vesicles with exudation and crusting

Subacute and chronic: dry, scaly, or excoriated papules or skin thickening (lichenification)

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Atopic Dermatitis



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In patients with highly pigmented skin:

- Dry skin may appear whitish or gray
- Erythema may appear violaceous or may not be visible at all
- Lichenified areas may appear hyperpigmented
- Post-inflammatory hyper- and hypopigmentation

Allergic Rhinitis

Inflammation of the nasal mucosa leading to paroxysms of sneezing, rhinorrhea, nasal obstruction; itching of the eyes, nose and palate; postnasal drip, cough, irritability, fatigue



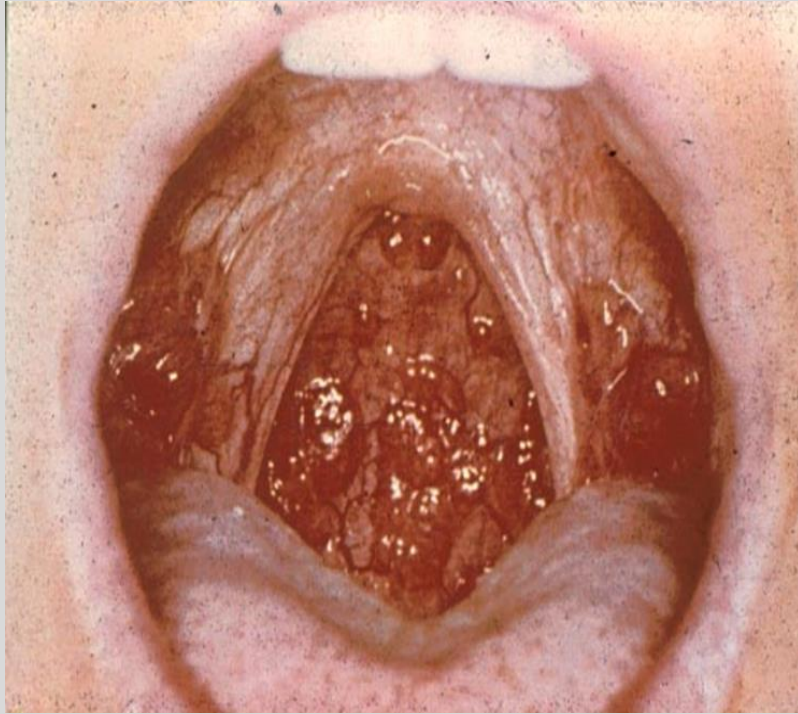
Allergic facies

Allergic salute



Allergic Rhinitis

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Lymphoid hyperplasia
“Cobblestoning”



Turbinate hypertrophy

Asthma

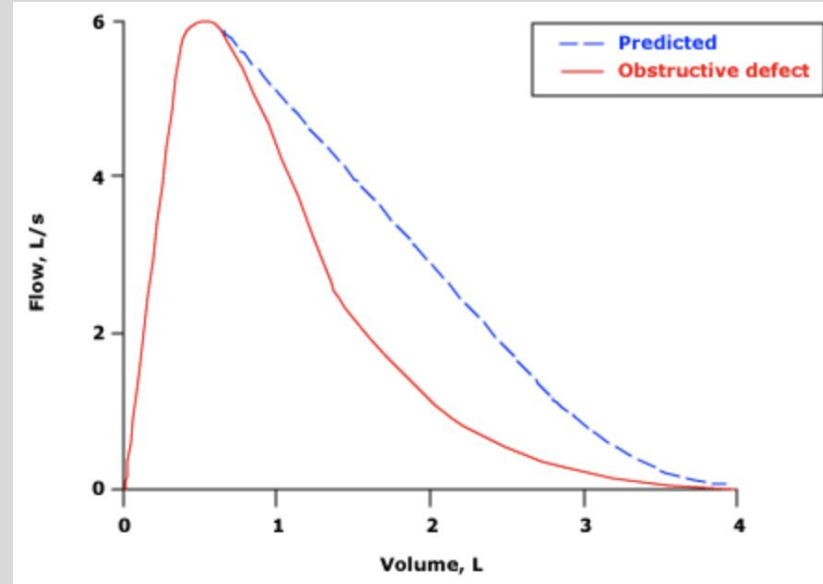
National Asthma Education and Prevention Program: common chronic disorder of the airways that is complex and characterized by variable and recurring symptoms, **airflow obstruction**, bronchial hyperresponsiveness, and an underlying **inflammation**.

Global Initiative for Asthma: a heterogeneous disease, usually characterized by chronic airway **inflammation**. It is defined by the history of respiratory symptoms such as wheeze, shortness of breath, chest tightness, and cough that vary over time and in intensity, together with variable **expiratory airflow limitation**.

Asthma

Four classic symptoms:

1. Wheezing
2. Cough (worse at night)
3. Shortness of breath,
dyspnea
4. Chest tightness



Assessment

Skin Patch/Skin Prick Testing



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Intradermal Testing

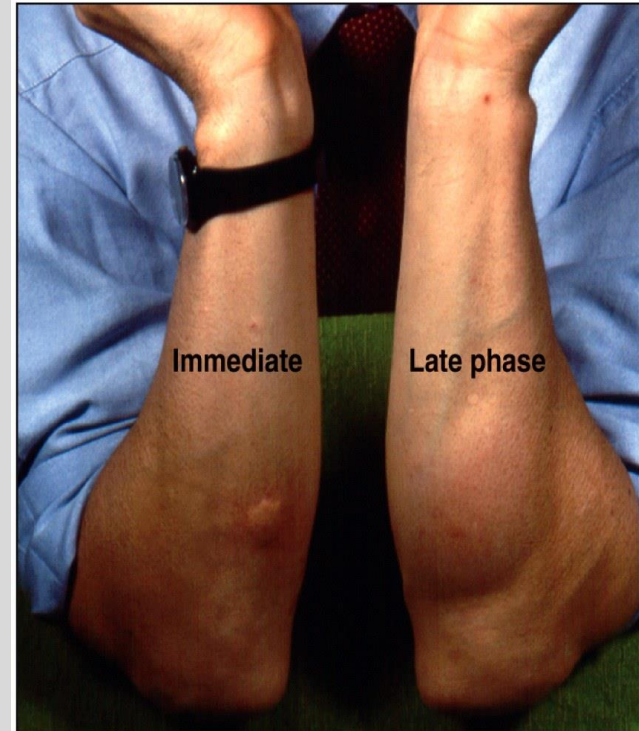
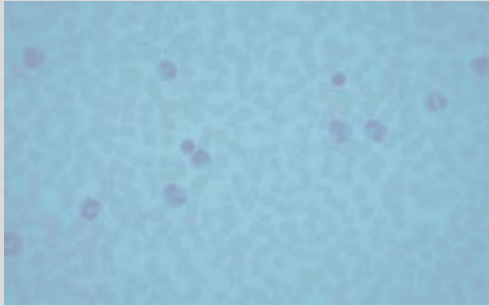


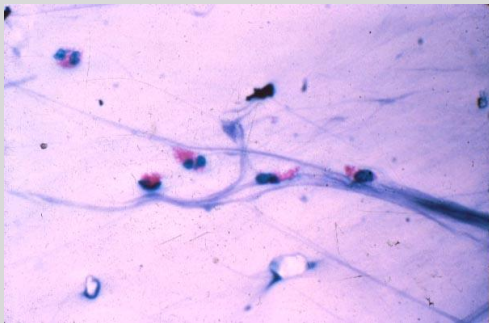
Figure 12.16 The Immune System, 3ed. (© Garland Science 2009)

Assessment

Eosinophils in blood



Eosinophils in sputum



Radioallergosorbent Testing

*-Allergen Specific IgE Antibody Testing (Blood)

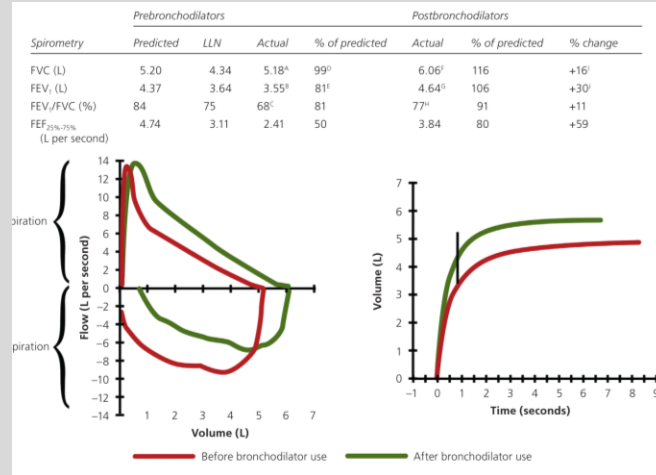
A 1 1 1

EGGWHITE (f1)	14.50	kUA/L	III+*
MILK (f2)	8.92	kUA/L	III+
FISH, COD (f3)	0.55	kUA/L	I+
WHEAT (f4)	>100	kUA/L	VI+
OAT (f7)	37.30	kUA/L	IV+
L SESAME SEED (f10)	3.65	kUA/L	III+
PEANUTS (f13)	21.30	kUA/L	IV+
SOYBEAN (f14)	2.87	kUA/L	II+
HAZELNUT (f17)	5.49	kUA/L	II+
ALMOND (f20)	3.45	kUA/L	II+
SHRIMP (f24)	22.00	kUA/L	r.v+
STRAWBERRY (f44)	2.05	kUA/L	II+
AVOCADO (f96)	1.35	kUA/L	II+
APPLE (f49)	2.60	kUA/L	II+
EGG YOLK (f75)	5.59	kUA/L	III+
L CHICKEN MEAT (f83)	<0.35	kUA/L	0
BANANA (f92) PECAN	1.16	kUA/L	II+
NET (f201) CASHEW	<0.35	kUA/L	0
NUT (f202)	13.90	kUA/L	III+
PISTACHIO (f203)	15.00	kUA/L	III+
WALNUT (f256)	3.13	kUA/L	II+
TURKEY MEAT (f284)	<0.35	kUA/L	0
MOLD (CLADOSPORIUM) (m2)	<0.35	kUA/L	0
MOLD (ASPERGILLUS) (m3)	<0.35	kUA/L	0
MOLD (FUSARIUM m9)	<0.35	kUA/L	0
ALLERGEN CONCENTRATION	SEE NOTE		
ALLERGEN	SEE NOTE		

SEE NOTE (QNS)

--- LEGEND
*--ABNORMAL

Pulmonary Function Testing



Case Study: Jack

A 4-year-old boy named Jack has been brought into the ED with swelling of the face, generalized itchiness, and drowsiness. The symptoms developed a few minutes after eating a chocolate bar. His mother explains that he has had minor reactions to pistachios in the past, including lip swelling. The boy also has eczema, and his mother thinks that in the past, this has flared up when he has eaten hazelnuts. She has concluded that he might be allergic to tree nuts and has tried to eliminate them from the house. The patient has never had any symptoms from milk. Jack asks if you're going to take away the itch, and if he can have more chocolate when he gets home.

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Summary

- Type I/Anaphylaxis/Atopic Hypersensitivity reactions are caused by degranulation of mast cells and eosinophils
- Type II/Antibody-mediated Hypersensitivity reactions are caused by IgG reacting with complement or Fc receptor on macrophages
- Type III/Immune complex Hypersensitivity reactions are caused by immune complexes of antigen and antibody that cause damages at the site of production (Arthus) or circulate and cause damage elsewhere (serum sickness)
- Type IV/Delayed Hypersensitivity reactions are caused directly by CD8+ cytotoxic T-cells or indirectly by effector CD4+ T-cells which initiate release of inflammation-inducing cytokines

Summary (cont'd)

- Anaphylaxis is a serious hypersensitivity reaction that can progress rapidly and may cause death.
- Epinephrine is the first line treatment for anaphylaxis and should be administered as early as possible if anaphylaxis is suspected.
- Atopy is a genetic propensity to produce allergen-specific IgE.
- Allergy is diagnosed when individuals have both allergen-specific IgE and symptoms upon exposure to that allergen.
- Atopic march/allergic march/allergic triad: atopic dermatitis + allergic rhinitis + asthma
- Diagnosis of hypersensitivity can consist of intradermal testing, patch/prick testing, blood testing, and/or pulmonary function testing.

	Type I: Immediate Hypersensitivity	Type II: Bound Antigen	Type III: Immune Complex	Type IV: Delayed Hypersensitivity
Onset	Seconds, if IgE is preformed	Seconds, if IgG is preformed	Hours, if IgG is preformed	2–3 days
Infectious trigger	Schistosomiasis	Immune hemolytic anemias	Poststreptococcal glomerulonephritis	Hepatitis B virus
Environmental trigger	House dust mite, peanut	Immune hemolytic anemias	Farmer's lung	Contact dermatitis
Autoimmunity	Not applicable	Immune hemolytic anemias	Systemic lupus erythematosus	IDDM, celiac disease, MS, RA
Adaptive immune system mediators	IgE	IgG	IgG	T cells
Innate immune system mediators	Mast cells, eosinophils	Complement, phagocytes	Complement neutrophils	Macrophages

IDDM, insulin-dependent diabetes mellitus; *Ig*, immunoglobulin; *MS*, multiple sclerosis; *RA*, rheumatoid arthritis.

Question #1

A 34-year-old woman is brought to the emergency department via EMS after collapsing at a work picnic in the park. She reports diffuse, crampy abdominal pain with one episode of vomiting on the way to the hospital. Temperature is 99.1F, pulse is 121 bpm, and blood pressure is 88/75mmHg. Oxygen saturation is 85% on room air. She is using accessory muscles to breathe and has high-pitched sounds on inspiration, and she has swelling around the eyes. Which of the following is most directly involved in causing this patient's symptoms?

- A. Bradykinin excess
- B. Histamine
- C. IgG autoantibodies
- D. Immune complexes
- E. Natural killer cells
- F. Th1 lymphocytes

Question #1

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- A. Bradykinin excess
- B. Histamine
- C. IgG autoantibodies
- D. Immune complexes
- E. Natural killer cells
- F. Th1 lymphocytes

Common presentations of anaphylaxis include hypotension, hives, pruritic and/or flushed skin, swollen lips or tongue, periorbital edema, and conjunctival swelling. Anaphylaxis is a Type I hypersensitivity reaction characterized by binding of antigen to bound IgE on mast cells and basophils, followed by degranulation of mast cells and release of vasoactive amines like histamine and serotonin.

Question #2

A 65-year-old male comes to the family medicine clinic for evaluation of pruritic, erythematous papules, vesicles and blisters in a linear and geometric pattern on the lower extremities that appeared two days ago. He reports that he has been wearing shorts for the past week now that the weather is warmer outside. His daily activities include a 3-mile morning walk with his wife, reading, gardening, and playing chess with his friends. He denies use of new soaps or lotions, recent travel, and sick contacts. Which of the following mechanisms is most important in the pathogenesis of this patient's most likely condition?

- A. Activation of sensitized B lymphocytes
- B. Antibodies binding to cell surface antigens
- C. Antibody-antigen complex deposition
- D. Differentiation of CD4+ T lymphocytes into helper T lymphocytes
- E. Leukotriene-mediated inflammation



Question #2

Which of the following mechanisms is most important in the pathogenesis of this patient's most likely condition?

- A. Activation of sensitized B lymphocytes
- B. Antibodies binding to cell surface antigens
- C. Antibody-antigen complex deposition
- D. Differentiation of CD4+ T lymphocytes into helper T lymphocytes
- E. Leukotriene-mediated inflammation

This patient is most likely experiencing an allergic contact dermatitis due to poison ivy, which he likely contacted while gardening. The allergic response to poison ivy is a type IV hypersensitivity reaction which requires secondary exposure to the antigen, which triggers differentiation of CD4+ T lymphocytes into T-helper type lymphocytes.

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THANK YOU!

If you have any questions, please email me at lrudin@nyit.edu

Evaluations:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448F>
[WYXREL4R](#)

Do.
Make.
Innovate.
Reinvent the Future.